

Fig. 4: An actual-size PCB layout for the circuit

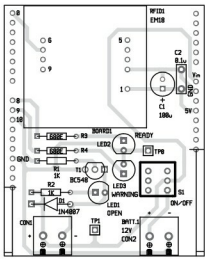


Fig. 5: Component layout for the PCB

ule from the circuit. For programming:

1. Open the AccessControl.ino sketch from the Arduino IDE
2. Change the tag ID in AccessControl.ino sketch with the ID you have noted down earlier
3. Connect the Arduino board (Board1) with the PC

4. Upload the sketch into the board

If uploading is successful, you will see the glowing of LED2. It means the system is ready to read the tag. Now, bring the tag near RFID1 reader. If tag ID matches with the ID in the code, solenoid lock will open for five seconds. It closes automatically after five

EFY Note

The source code of this project is included in this month's EFY DVD and is also available for free download at source.efymag.com

seconds. Glowing of LED1 indicates that the lock is open. Glowing of warning LED3 means that you are using the wrong tag.

Construction and testing

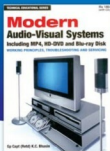
An actual-size, single-side PCB (Arduino shield type) for access control is shown in Fig. 4 and its component layout in Fig. 5. Assemble the circuit on recommended PCB to save time and minimise assembly errors. Double-check for any overlooked error.

To test the circuit for proper functioning, verify correct 12V supply for the circuit at TP1 with respect to TP0. ●

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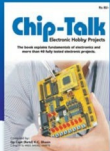
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